

Astro 142

Discussion 7

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Importing astro databases with numpy/pandas

Astroquery: <https://astroquery.readthedocs.io/en/latest/index.html>

Results are (usually) given as an [astropy table](#), similar in concept to a pandas dataframe. [See here for info on switching between pandas/astropy tables](#)

Some databases include options to get data directly off a server and into python (usually in the [hdulist format that's used for fits images](#))

SQL queries

SQL - structured query language. Used to request/modify specific parts of (usually very large) databases when working with the entire database is unnecessary/impractical

Lots of tutorials online, e.g. [this one](#)

Often use modified versions of SQL (e.g. [ADQL for Vizier](#) or [for SIMBAD](#)) with similar syntax

Many databases have unique syntax (e.g. [ADS](#)), but similar principles

Discussion Activity

Discussion 7 Activity

1. Go to the Vizier ADQL page and load the YSO demo: <http://tapvizier.cds.unistra.fr/adql/>. On a slide, list what you think the column names of the resulting CSV file will be, then screenshot the actual column names.
2. Install the pyvo and astroquery packages (you'll need these for homework)
3. Use astroquery (<https://astroquery.readthedocs.io/en/latest/simbad/simbad.html>) to get the coordinates for M1 from Simbad and screenshot on a slide
4. Access the nearby galaxy catalog from this paper in Vizier. Use astroquery (<https://astroquery.readthedocs.io/en/stable/vizier/vizier.html>) to get the name, ra, dec, distance, radial velocity from Vizier. Attach screenshots on the next slides.
<https://ui.adsabs.harvard.edu/abs/2013AJ...145..101K/abstract>

Query for M1

```
dyson@dpl-linux:~$ python3
Python 3.10.12 (main, Aug 15 2025, 14:32:43) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> from astroquery.simbad import Simbad
>>> table = Simbad.query_object('m1')
>>> print(table)
```

main_id	ra	dec	...	coo_wavelength	coo_bibcode	matched_id
	deg	deg	...			
M	1	83.6324	22.0174	...	X 2022A&A...661A..38P	M 1

```

Python 3.10.12 (main, Aug 15 2025, 14:32:43) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> from astroquery.vizier import Vizier
>>> vizier = Vizier()
>>> list = vizier.find_catalogs('Updated Nearby Galaxy Catalog')
>>> for k, v in list.items():
...     print(k, ":", v.description)
...
J/ApJS/220/6 : Updated catalog of GALEX nearby galaxies (Bai+, 2015)
J/AJ/145/101 : Updated nearby galaxy catalog (Karachentsev+, 2013)
>>> vizier.ROW_LIMIT = 2
>>> galaxy = vizier.get_catalogs("J/AJ/145/101")
>>> galaxy
TableList with 6 tables:
  '0:J/AJ/145/101/catalog' with 29 column(s) and 2 row(s)
  '1:J/AJ/145/101/table3' with 7 column(s) and 2 row(s)
  '2:J/AJ/145/101/table4' with 5 column(s) and 2 row(s)
  '3:J/AJ/145/101/table5' with 5 column(s) and 2 row(s)
  '4:J/AJ/145/101/table6' with 6 column(s) and 2 row(s)
  '5:J/AJ/145/101/refs' with 4 column(s) and 2 row(s)
>>> galaxy[0]
<Table length=2>
  Name      Nd   RAJ2000   DEJ2000   ...   Ti1      K+04 SimbadName  NEDname
  str18     int16   str10      str9      ... float32 int16   str29      str29
  -----
  UGC12894    7 00 00 22.5 +39 29 44 ...   -1.5      0   UGC12894   UGC12894
    WLM       8 00 01 58.1 -15 27 40 ...    0.0      1   WLM GALAXY WLM GALAXY
>>> galaxy[1]
<Table length=2>
  Name      C   l_mag   mag   e_mag   r_mag Flt
  str18     uint8  str1 float32 float32 str5 str3
  -----
  UGC12894    1      17.57   0.10 CW12a FUV
  UGC12894    1      16.80    -- CW11a  B
>>> galaxy[2]
<Table length=2>
  Name      C   HRV   e_HRV   r_HRV
                km / s km / s

```


